1.7 Types of Functions

In this section, we will discuss the following types of functions with suitable examples.

(i) one – one (ii) many – one (iii) onto (iv) into

1.7.1 One – one function

Let us assume that we have a cell phone with proper working condition. If you make a usual call to your friend then you can make only one call at a time (Fig.1.21).

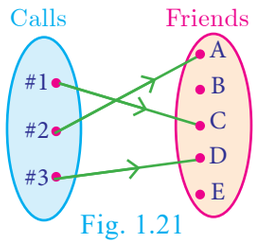


Fig. 1.21

If we treat making calls as a function, then it will be one - one.

A function $f : A \rightarrow B$ is called one – one function if distinct elements of A have distinct images in B.

A one-one function is also called an **injection**.

Equivalently,

If for all $a_1, a_2 \in A$, $f(a_1) = f(a_2)$ implies $a_1 = a_2$, then f is called one – one function.

Illustration 10

$A = \{1,2,3,4\}$ and $B = \{a,b,c,d,e\}$

(i) Let $f = \{(1,a), (2,b), (3,d), (4,c)\}$

In Fig. 1.22, for different elements in A, there are different images in B.

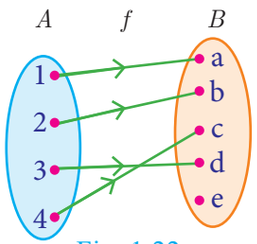


Fig. 1.22

Hence f is a one – one function.

(ii) Let $g = \{(1,b), (2,b), (3,c), (4,e)\}$

g is a function from A to B such that $g(1) = g(2) = b$, but $1 \neq 2$. Thus two distinct elements 1 and 2 in the first set A have same image b the second set in B (Fig.1.23). Hence, g is not a one–one function.

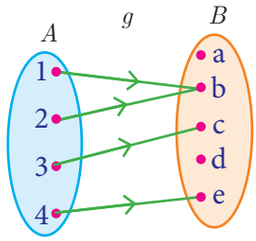


Fig. 1.23

1.7.2 Many – one function

In a theatre complex three films F_1, F_2, F_3 are shown. Seven persons (P_1 to P_7) arrive at the theatre and buy tickets as shown (Fig.1.24).

If the selection of films is considered as a relation, then this is a function which is many–one, since more than one person may choose to watch the same film.

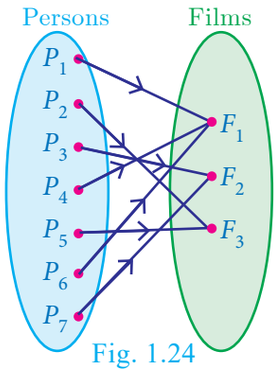


Fig. 1.24

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